

Bernardo Bahia Monteiro

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EDUCATION

University of Michigan

PhD Aerospace Engineering — GPA: 4.0

Advisors: Carlos E. S. Cesnik, Ilya Kolmanovsky

Research interest: Develop control-related metrics for Multidisciplinary Design Optimization (MDO)

that can estimate the closed-loop performance of the aircraft without control design

Projects: Airbus-Michigan Center for Aero-Servo-Elasticity of Very Flexible Aircraft (CASE-VFA, 2019–2022)

NASA's Supersonic Configurations at Low Speeds (SCALOS, 2022–present)

Ann Arbor, MI

September 2019–April 2024

(expected)

Universidade Federal de Minas Gerais

BSc Aerospace Engineering — GPA: 4.9/5.0

Admitted in first place; graduated first in class

Research assistant under Prof. Ricardo Utsch (2015–2018, numerical analysis)

Model rocketry team

Belo Horizonte, Brazil

February 2014–June 2019

University of Strathclyde

One semester exchange in Aero-mechanical Engineering

First class result in all disciplines, mostly at the master's level

Summer research assistant for the “Future UK Small Payload Launcher” project:

Feasibility study of a horizontal rocket launcher with a reusable first stage

Multidisciplinary Design Optimization and Optimal Control

Glasgow, UK

January–July, 2017

WORK EXPERIENCE

Embraer

Flight Operations Engineering

Design and implement software solutions to meet customers' demands (Java/C)

e.g., calculate friction coefficient of a contaminated runway from flight data

Support meetings with airline customers

Formulate and develop proof-of-concept solutions for flight data analysis (Python)

Work in an agile development team

São José dos Campos, Brazil

January, 2018–January, 2019

Preliminary Design

Program semi-empirical aerodynamic models adapted for the in-house MDO code (Fortran)

Attend design reviews

February, 2019–June, 2019

PUBLICATIONS

- [1] Bahia Monteiro, B., Gray, A. C., Cesnik, C. E. S., Kolmanovsky, I., and Vetrano, F., “Bi-level Multidisciplinary Design Optimization of a Wing Considering Maneuver Load Alleviation and Flutter,” *AIAA SCITECH 2023 Forum*, 2023, p. 0728.
- [2] Bahia Monteiro, B., Cesnik, C., and Kolmanovsky, I., “Gust Load Alleviation Control Related Metric for Multidisciplinary Design Optimization,” *International Forum on Aeroelasticity and Structural Dynamics*, June 2022.
- [3] Jonsson, E., Riso, C., Bahia Monteiro, B., Gray, A. C., Martins, J. R. R. A., and Cesnik, C. E. S., “High-Fidelity Gradient-Based Wing Structural Optimization Including a Geometrically Nonlinear Flutter Constraint,” *AIAA SciTech Forum*, Jan. 2022.

HONOURS AND AWARDS

Gold medal award for best GPA of the class of 2019 at UFMG

2nd overall place on ENEM 2013 (Brazilian version of the SAT) — more than 7 million students applied

Gold medal at the XV Brazilian Olympiad of Astronomy and Astronautics (October 2012)

Honorable mention on the state's Olympiad of Mathematics (MG, 2007)

COMPUTER SKILLS

C/C++, Python, Java, Fortran, Git, Matlab/Simulink, NASA's OpenMDAO, Linux/Unix (real-time), L^AT_EX

LANGUAGE SKILLS

Portuguese (native), English (near-native), Spanish, French, German (beginner)

HOBBIES

Mountain biking, tennis, gliding, road cycling, skiing, climbing, sculling